

The Text discusses the Cost of poor Software Quality (CPSQ) in the United States in 2020. As Software become increasingly important for businesses and society, organizations need to deliver products quickly while maintaining quality and security. However, many companies prioritize speed over software quality, which creates significant financial losses.

The report estimates that poor software quality cost the US approximately \$2.08 trillion in 2020. The main causes were operational software failures, unsuccessful IT projects, and problems with legacy systems. Operational failures were the largest contributor, costing around \$1.56 trillion, while unsuccessful projects cost approximately \$260 billion and legacy systems around \$520 ^{Billion} trillion.

The report also highlights the growing importance of cybersecurity and technical debt. Vulnerable software can lead to cyberattacks, data breaches, ransomware, and other serious problems. Technical debt was estimated at \$636 trillion, representing the future cost of fixing existing software problems.

The authors argue that organizations should focus more on preventing defects instead of fixing them after they cause damage. They recommended secure coding, code analysis, quality measurement, better project management, evaluation of third-party components, and improved maintainability of legacy systems.

Additionally, the report explains that software development is becoming increasingly difficult because systems are growing in size and complexity, technologies are constantly changing, and companies face increasing pressure to deliver products faster. The growth of cloud computing, AI, IoT, automation, and digital services also creates more potential points of failure and security vulnerabilities.

The Covid-19 pandemic further accelerated digital transformation, increasing the demand for remote work, online education, cloud services, and digital products. As a result, society became even more dependent on software, making software failures more serious and potentially more damaging.

The report concludes that organizations need to treat software development more like a true engineering discipline, where quality, security, and risk prevention are considered throughout the entire development lifecycle. Overall, the main message is that investing in software quality can reduce costs, improve security, encourage innovation, and lead to more successful projects.

Finally, the report shows that high-performing software organizations can achieve much better results by following strong development and quality practices. Important factors include good processes, accurate estimation, skilled professionals, defect prevention, customer focus, and a strong culture of quality.